u_t, $\sigma_\mu^2 = \text{var}(e_t^p)/\text{var}(z_t)$ and $\sigma_M^2 = \text{var}(e_t^y)/\text{var}(z_t)$, and these asymptotic standard errors are those that an OLS procedure would report that did not account for the fact that the values of m^y and $\check{\eta}_t(m^y)$ were estimated. In addition, the optimal Γ remains as in proposition 3.

The fact that the standard errors are correct is a bit surprising at first, as z_t is a generated regressor. When there are no other controls, $C_t^p = C_t^y = 0$, the reason is that the GIV is directly obtained from an exact formula ($z_t := S'Qy_t$ as in [31]) and can thus be constructed without error.[14] For this result, assumption 1 and the fact that we know the σ_{u_i} are important. If one needs to estimate λ or the σ_{u_i}'s, then that estimation error needs to be taken into account when calculating the standard errors of the elasticities, ψ and ϕ^d, and pass-throughs, M and μ (see secs. IV.B and IV.C).

III. Discussion

A. *The Intuition Behind GIV Estimators*

1. GIVs in a Simple Model

To explain the intuition, we specialize our analysis to the case where there is only a single factor and all entities have the same loading on the factor, $\lambda_i = 1$, and there are no other controls, $C_t^p = C_t^y = 0$. The single factor is then absorbed by a time fixed effect. It allows us to develop the main intuition in a transparent way. The system is

$$p_t = \psi y_{St} + \varepsilon_t, \tag{38}$$

$$y_{it} = \phi^d p_t + \eta_t + u_{it}. \tag{39}$$

To take advantage of the great analytical simplicity of that example, we retrace the derivation steps in an elementary manner. We cannot estimate ψ and ϕ^d by OLS as ε_t and η_t are typically correlated, implying that y_{St} is correlated with ε_t in (38) and p_t with η_t in (39).

We construct the GIV as the "size-weighted" average outcome, $y_{St} = \Sigma_i S_i y_{it}$, minus the "equal-weighted" average outcome, $y_{Et} = (1/N)\Sigma_{i=1}^N y_{it}$:

$$z_t := y_{\Gamma t} = y_{St} - y_{Et}. \tag{40}$$

Given that[15]

[14] When there is an m^y, the reason is more subtle and revealed by the proof: an error in m^y creates an error proportional to C_t^y times random variables uncorrelated to it, hence a negligible term $o_p(T^{-1/2})$.

[15] As is clear in this simple example, any average (other than size weighted) of the y_{it} can be used to remove $\phi^d p_t + \eta_t$, not only y_{Et}. What is most efficient, however, is to use the equal-weighted average in this case.

$$y_{St} = \phi^d p_t + \eta_t + u_{St}, \qquad y_{Et} = \phi^d p_t + \eta_t + u_{Et}, \tag{41}$$

the GIV is also

$$z_t = u_{\Gamma t} = u_{St} - u_{Et}, \tag{42}$$

and it is made only of idiosyncratic shocks. Given the exogeneity condition (3), we have $\mathbb{E}[u_t \varepsilon_t] = 0$, and hence $\mathbb{E}[z_t \varepsilon_t] = 0$. Using (38), this gives $\mathbb{E}[(p_t - \psi y_{St}) z_t] = 0$, and thus $\psi = \mathbb{E}[p_t z_t] / \mathbb{E}[y_{St} z_t]$. Its empirical counterpart is the estimator $\psi_T^e = \Sigma_{t=1}^T p_t z_t / \Sigma_{t=1}^T y_{St} z_t$.

Likewise, given the exogeneity condition (3), we have $\mathbb{E}[z_t \eta_t] = 0$. In addition, under assumption 3 that the u_{it} are homoskedastic we have $\mathbb{E}[u_{Et} u_{\Gamma t}] = 0$.[16] So, using (41), we have $\mathbb{E}[(y_{Et} - \phi^d p_t) z_t] = 0$ and hence $\phi^d = \mathbb{E}[y_{Et} z_t] / \mathbb{E}[p_t z_t]$. Its empirical counterpart is $\phi_T^{d,e} = \Sigma_{t=1}^T y_{Et} z_t / \Sigma_{t=1}^T p_t z_t$.

Hence, the same instrument—the GIV z_t—can be used to estimate both the demand elasticity $\phi^d = \mathbb{E}[y_{Et} z_t] / \mathbb{E}[p_t z_t]$ and the supply elasticity $\phi^s = 1/\psi = \mathbb{E}[y_{St} z_t] / \mathbb{E}[p_t z_t]$. Intuitively, an idiosyncratic shock to, for instance, a large country affects both world prices and quantities, so it allows us to estimate the elasticity of demand of the other countries and the elasticity of supply, which is equal (as supply equals demand) to the size-weighted demand of all countries.[17]

The general propositions 1 and 2 show that those estimators are $\sqrt{T}$-consistent. Proposition 3 implies that this GIV (40) is the optimal one, as stated below.

COROLLARY 1 (GIV estimator in the case with a single factor and common loadings). Let assumption 3 hold. In the case with a single factor and common loadings, (38)–(39), the optimal GIV estimator of proposition 3 takes the form $\Gamma = S - E$—that is, $\Gamma_i = S_i - (1/N)$:

$$z_t := y_{\Gamma t} = y_{St} - y_{Et}, \tag{43}$$

so that $z_t = u_{\Gamma t} = u_{St} - u_{Et}$.

Hence, in this simplest of cases, the GIV is the "size-weighted" minus the "equal-weighted" value of outcomes, which recovers the "size-weighted" minus the "equal-weighted" value of idiosyncratic shocks. In addition, $\tilde{u}_{it} = u_{it} - u_{Et}$, so that $z_t = \Sigma_i S_i \tilde{u}_{it}$. In the general case, intuitively, the GIV weights Γ mimic this structure, but they need to be adjusted to remove common factors if the factor structure is more complex; see (17).

[16] Indeed, we have $\Gamma' E = \Sigma_i [S_i - (1/N)](1/N) = 0$. Thus, $\mathbb{E}[u_{\Gamma t} u_{Et}] = \mathbb{E}[(\Gamma' u_t)(u_t' E)] = \Gamma' \mathbb{E}[u_t u_t'] E = \sigma_u^2 \Gamma' E = 0$.

[17] As the supply elasticity $\phi^s = \mathbb{E}[y_{St} z_t] / \mathbb{E}[p_t z_t]$ traces out how the supply side reacts to the price, we need to use aggregate supply y_{St}, which is, in equilibrium, the size-weighted average of the demands. Alternatively, the demand elasticity $\phi^d = \mathbb{E}[y_{\tilde{E}t} z_t] / \mathbb{E}[p_t z_t]$ indicates how the demand side of an individual country reacts to the price, so one can take many weights $\tilde{E}$, provided that they satisfy (10). It is most efficient, however, to take the equal-weighted average of demands of individual countries, $\tilde{E}_i = E_i = 1/N$, to maximally smooth out idiosyncratic noise.

2. GIV Estimators Are More Precise in Concentrated Economies with Volatile Idiosyncratic Shocks

We next study when the GIV achieves a good precision.

COROLLARY 2 (Precision of the GIV estimator in the case with a single factor and common loadings). Let assumption 3 hold. Consider the case with a single factor and common loadings, (38)–(39). The GIV estimators are consistent and converge as $\sqrt{T}(\psi_T^e - \psi) \xrightarrow{d} N(0, \sigma_\psi^2)$ and (assuming that $\psi \neq 0$) $\sqrt{T}(\phi_T^{d,e} - \phi^d) \xrightarrow{d} N(0, \sigma_{\phi^d}^2)$, where, assuming that ε_t and $\varepsilon_t^\phi := \eta_t + u_{Et}$ are homoskedastic conditionally on u_t, the asymptotic standard deviations of the scaled and centered GIV estimators are

$$\sigma_z = \sigma_{u_G} = h\sigma_u, \qquad \sigma_\psi = \frac{\sigma_\varepsilon}{\sigma_z|M|}, \qquad \sigma_{\phi^d} = \frac{\sigma_{\varepsilon^\phi}}{\sigma_z|\mu|}, \tag{44}$$

where h represents the excess Herfindahl:

$$h := \sqrt{\left(\sum_{i=1}^{N} S_i^2\right) - \frac{1}{N}}. \tag{45}$$

Corollary 2 highlights what it takes to obtain a strong instrument and a precise estimate of ψ (and similarly of ϕ^d): we need one or several large units (to have a large excess Herfindahl h), and we need idiosyncratic shocks to be volatile compared with the volatility of aggregate shocks (large σ_u compared with $\sigma_{\varepsilon^\psi}$ or $\sigma_{\varepsilon^\phi}$).[18]

B. *Instrument Strength*

Two-stage least squares (2SLS).—One can view regression (36) as the first-stage regression of the price on the GIV and controls c_t, which yields the instrumented price $p_t^e := \mu^e z_t + b^p c_t$. In the second stage, we regress supply on the instrumented price and controls c_t:

$$y_{St} = \frac{1}{\psi} p_t^e + \beta^{y_S} c_t + \varepsilon_t^{y_S,\text{2SLS}}, \tag{46}$$

which gives an estimate of the supply elasticity $1/\psi$. To estimate the demand elasticity, we regress the equal-weighted (not size-weighted) demand on the instrumented price, p_t^e, and controls c_t in the second stage: $y_{\bar{E}t} = \phi^d p_t^e + \beta^{y_{\bar{E}}} c_t + \varepsilon_t^{y_{\bar{E}},\text{2SLS}}$. This yields estimates that are the same as the IV estimators from proposition 4.

GIVs and weak instruments.—In the case of the OLS estimator, a low power of the GIV simply manifests itself as large standard errors, while

[18] This intuition extends to the model with a more complex factor structure: the expression $\sigma_z = h\sigma_u$ now entails a modified Herfindhal $h = \sqrt{S'QS}$, which is less transparent (see proposition 3) but has broadly similar behavior. For an example, see the proof of corollary 2.

in the case of the IV estimators, we can encounter weak instrument problems. For the parameters that can be estimated using OLS—that is, μ and M—the standard errors obtained via standard OLS inference are valid when T is large (as per proposition 5 and under its assumptions). When a ratio is implicitly performed—for instance, to estimate ϕ^d or ψ by instrumental variables—the 2SLS procedure will also give correct standard errors when the instrument is strong enough. A traditional rule of thumb for the strength of the instrument (in the i.i.d., homoskedastic case) is that the F-statistic (which is the squared t-statistic on μ) on the first stage (37) should be greater than a threshold around 16–19, and this advice is being progressively enhanced in current IV research (see Montiel Olea and Pflueger 2013; Andrews, Stock, and Sun 2019).

Proposition 3 and corollary 2 show that we have a precise estimator if concentration is high and if the idiosyncratic shocks are volatile relative to the volatility of aggregate shocks. Hence, before conducting a large-scale study and data collection effort, researchers can perform an *ex ante power analysis*. Recall that in the most basic case, the standard error of the estimator is $\mathrm{SE}(\psi_T^e) = \sigma_\varepsilon/(h\sigma_u|M|\sqrt{T})$ (see [44]). To get a quick sense of the power of GIVs, we can have a rough estimate of σ_ε using the volatility of the left-hand-side variable (here, p_t) as an order of magnitude, σ_u, using $\sigma_{\tilde{y}_i}$ so that a simple common factor is removed as well as the component that depends on prices, and the average excess Herfindahl h. These inputs can be used to calibrate an order of magnitude for $\sigma_{\psi_T^e}$, given the researcher's previous view on M. This gives a sense of the t-statistics $t = \psi/\sigma_{\psi_T^e}$ that can be obtained with T periods of data. Hence, before refining the empirical model and perhaps collecting additional data, one has a sense of whether the GIV will be sufficiently powerful. The results of such a power analysis can be reported alongside the final estimates.

C. *Robustness to Misspecification and Threats to Identification*

Before discussing the limitations of GIVs and diagnostics for misspecification, we highlight the forms of misspecification to which GIV estimators are robust.

Forms of misspecification to which GIVs are robust.—First, it is possible to construct GIVs based on a subset of entities, I_t—that is, $z_t = \sum_{i \in I_t} S_i \check{u}_{it}$. This can be useful in practice as we can select the top K entities, select the entities for which we have data, or omit entities for which the data may contain measurement error. In this case, all results go through, although a rescaling may be required to ensure that (35) holds.[19] Hence, the estimator remains valid, although it is not the optimal GIV estimator.

[19] For instance, we still have $u_{St} = z_t + \varepsilon_t^{u_S}$, with $z_t \perp \varepsilon_t^{u_S}$. Appendix section D.10 gives a formal analysis.